University of Eloued

Series 08
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Destination: Pro.Atallah amore and Dr. Ghanaim Zouhira

K_means Clustering

K-Means clustering is an unsupervised machine learning algorithm that partitions a dataset into K distinct, non-overlapping groups by minimizing the distance between data points and their cluster's center (centroid). The algorithm works iteratively: it assigns each data point to the nearest centroid, then recalculates the centroids based on the mean of the assigned points, repeating until the clusters stabilize.

Objective Function The algorithm aims to minimize the Within-Cluster Sum of Squares (WCSS), which measures the total variance within all clusters:

$$j = \sum_{j=1}^{k} \sum_{x_i \epsilon\, C_j} (\|x_i - \mu_j\|)^2$$

- ✓ K is the total number of clusters.
- ✓ C_j: represents the j-th cluster.
- ✓ x_i:is an individual data point belonging to C_j:.
- ✓ μ_j: is the centroid (mean) of cluster C_j.
- ✓ $(\|x_j - \mu_j\|)^2$is the squared Euclidean distance between a data point and its centroid.

Centroid Update After points are assigned, each centroid is updated to be the exact center of its new cluster using the mean formula:

$$\mu_j = \frac{1}{C_j} \sum_{x_i \epsilon C_j} x_i$$

- $|C_j|$: is the total number of data points assigned to cluster $|C_j|$ ..

Exercise 2: Algorithmic Implementation of K-Means

Introduction: Automating E-Commerce and Marketing Analytics

In the digital retail and e-commerce sector, marketing analysts must process thousands of customer interactions daily to optimize advertising budgets and personalize the user experience. While our manual calculations proved the mathematical theory of K-Means, real-world business data science requires computational scale. In this exercise, you will step into the role of a commercial data analyst. By leveraging Python's machine learning libraries, you will automate the customer segmentation process to instantly uncover actionable marketing personas from our digital retail dataset.

Dataset:

Customer	X_1Age	X_2: Income (10k)	X_3: Visits/month	X_4 Clicks/visit	X_5: Spend (10)
A	20	3	10	15	2
B	25	4	8	12	3
C	50	9	2	3	10
D	55	10	1	2	12
E	45	8	3	5	9

Question 6: Python Implementation with Scikit-Learn

Having manually calculated the Expectation-Maximization cycle to understand the algorithm's internal geometry, your task is now to automate this exact process using Python.

Write a complete Python script using the `scikit-learn` library to apply K-Means clustering to our original 5-dimensional customer dataset. Your code must execute the following steps:

1. Import the required machine learning and data manipulation libraries.
2. Construct the dataset containing the five customers and their five variables (Age, Income, Visits, Clicks, Spend) using either a NumPy array or a Pandas DataFrame.

3. Initialize the K-Means model to define exactly two target clusters (K=2) and fit the model to your dataset.
4. Programmatically extract and print the final cluster assignment (label) for each customer.
5. Programmatically extract and print the final 5-dimensional coordinates of the two centroids to verify they match our manual convergence results.

Solution:
You find the full solution n the next Link:

https://github.com/atman500/atman500/blob/main/K_mean_Clustring_model.ipynb

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Second Example using

Use the same question of the previous Exercise

You find the full solution of the second example in the below Link

https://github.com/atman500/atman500/blob/main/k_mean_multiple.ipynb